

3. Inter-Processor Communication (IPC)

3.1 Overview

Inter-Processor Communication (IPC) supports hardware sharing and communication between two processors. IPC can generate interrupt events to support communication between each processor.

Several semaphore mechanisms are required to achieve mutually exclusive access between processors. To achieve this, IPC implements multiple locks. Application developers select to use separate or shared hardware semaphores between the two CPUs. A mechanism is also required for simple data communication between CPUs. For example, a FIFO that CPU0 uses for write-only data and CPU1 uses for read-only data. In addition, for mutual use, a FIFO is required with CPU1 as write-only data and CPU0 as read-only data. [Table 3.1](#) lists the IPC specifications.

Table 3.1 IPC specifications

Item		Description
Semaphore	Lock information	- A total of 16 factors - Security attributes and privileged attributes are selected every 8 factors
Inter-Processor interrupt	Non-maskable	- IPC0: one factor, one non-maskable interrupt, security attributes and privileged attributes are selected. - IPC1: one factor, one non-maskable interrupt, security attributes and privileged attributes are selected.
	Maskable	- IPC00: 8 factors, one maskable interrupt, security attributes and privileged attributes are selected*¹ - IPC01: 8 factors, one maskable interrupt, security attributes and privileged attributes are selected*¹ - IPC10: 8 factors, one maskable interrupt, security attributes and privileged attributes are selected*¹ - IPC11: 8 factors, one maskable interrupt, security attributes and privileged attributes are selected*¹
Message-FIFO	FIFO	- Message FIFO 00 (CPU1 -> CPU0), one maskable interrupt, security attributes and privileged attributes are selected*¹ - Message FIFO 01 (CPU1 -> CPU0), one maskable interrupt, security attributes and privileged attributes are selected*¹ - Message FIFO 10 (CPU0 -> CPU1), one maskable interrupt, security attributes and privileged attributes are selected*¹ - Message FIFO 11 (CPU0 -> CPU1), one maskable interrupt, security attributes and privileged attributes are selected*¹ 4 FIFO stages - Transfer data size 32 bits
	Error information	- Write with FIFO FULL*¹ - Read with FIFO Empty*¹

Note 1. The interrupts for IPC00, Message FIFO 00 and Error information are common. The same applies to others.

3.2 Register Descriptions

IPC_BASE (base address of IPC) is 0x4002_0000 (Secure) and 0x5002_0000 (Non-secure).

CPSCU_BASE (base address of CPSCU) is 0x4000_8000 (Secure) and 0x5000_8000 (Non-secure).

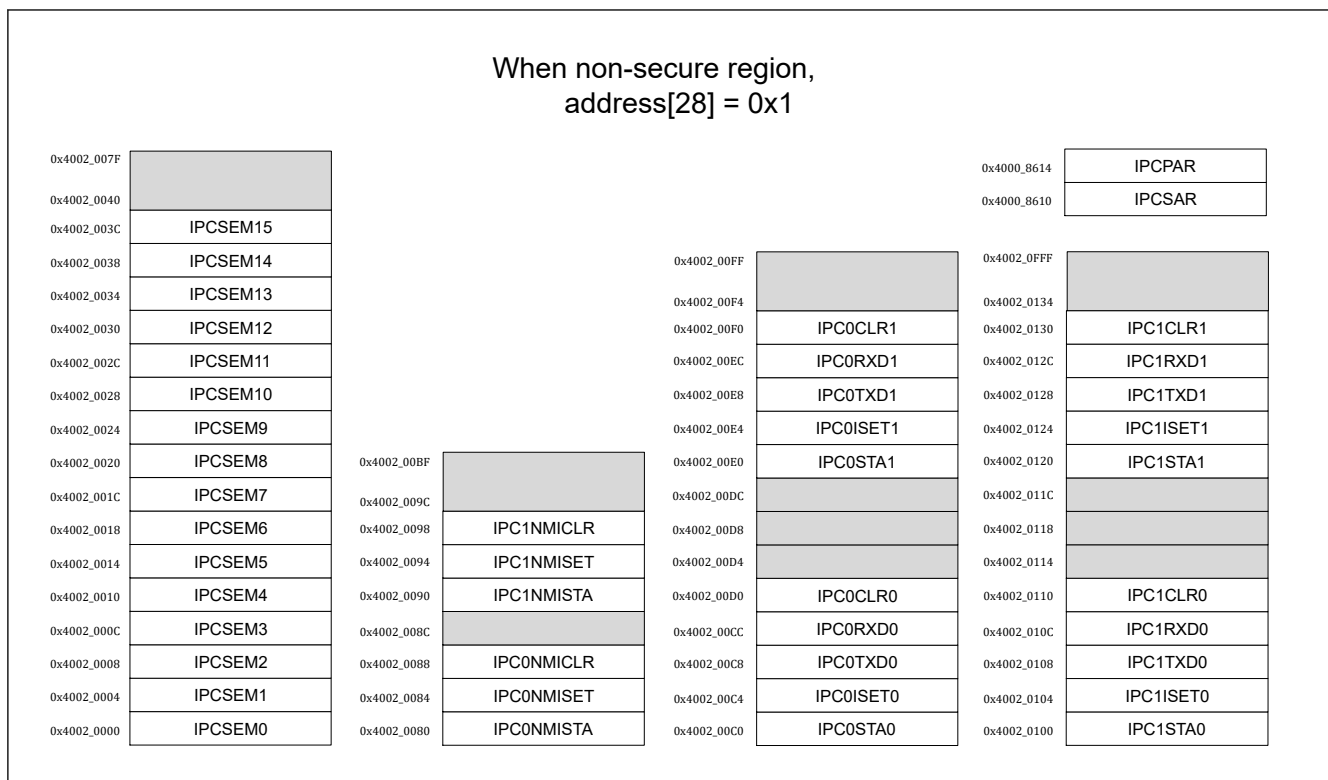


Figure 3.1 IPC register address map

3.2.1 IPCSAR : IPC Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x610

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SAIPC IR3	SAIPC IR2	SAIPC IR1	SAIPC IR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SAIPC NMI1	SAIPC NMI0	—	—	—	—	—	—	SAIPC SEM1	SAIPC SEM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SAIPCSEM0	Security attributes of registers for IPCSEMn (n = 0 to 7) Security attributes of IPCPAR.PAIPCSEM0 0: Secure 1: Non-secure	R/W
1	SAIPCSEM1	Security attributes of registers for IPCSEMn (n = 8 to 15) Security attributes of IPCPAR.PAIPCSEM1 0: Secure 1: Non-secure	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	SAIPCNMI0	Security attributes of IPC0NMI0, IPC0NMI1 and IPC0NMI2 Security attributes of IPCPAR.PAIPCNMI0 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
9	SAIPCNMI1	Security attributes of IPC1NMISTA, IPC1NMISSET and IPC1NMICLR Security attributes of IPCPAR.PAIPCNMI1 0: Secure 1: Non-secure	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	SAIPCIR0	Security attributes of registers for IPC0STA0, IPC0ISET0, IPC0TXD0, IPC0RXD0 and IPC0CLR0 Security attributes of IPCPAR.PAIPCIR0 0: Secure 1: Non-secure	R/W
17	SAIPCIR1	Security attributes of registers for IPC0STA1, IPC0ISET1, IPC0TXD1, IPC0RXD1 and IPC0CLR1 Security attributes of IPCPAR.PAIPCIR1 0: Secure 1: Non-secure	R/W
18	SAIPCIR2	Security attributes of registers for IPC1STA0, IPC1ISET0, IPC1TXD0, IPC1RXD0 and IPC1CLR0 Security attributes of IPCPAR.PAIPCIR2 0: Secure 1: Non-secure	R/W
19	SAIPCIR3	Security attributes of registers for IPC1STA1, IPC1ISET1, IPC1TXD1, IPC1RXD1 and IPC1CLR1 Security attributes of IPCPAR.PAIPCIR3 0: Secure 1: Non-secure	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

This register is write-protected by PRCR_S.PRC4.

SAIPCSEM0 bit

Security attributes of registers. The target registers are as follow.

- IPCSEM0 to IPCSEM7 registers
- IPCPAR.PAIPCSEM0 bit

SAIPCSEM1 bit

Security attributes of registers. The target registers are as follow.

- IPCSEM8 to IPCSEM15 registers
- IPCPAR.PAIPCSEM1 bit

SAIPCNMI0 bit

Security attributes of registers. The target registers are as follow.

- IPC0NMISTA register
- IPC0NMISSET register
- IPC0NMICLR register
- IPCPAR.PAIPCNMI0 bit

SAIPCNMI1 bit

Security attributes of registers. The target registers are as follow.

- IPC1NMISTA register
- IPC1NMISSET register
- IPC1NMICLR register
- IPCPAR.PAIPCNMI1 bit

SAIPCIR0 bit

Security attributes of registers. The target registers are as follow.

- IPC0STA0 register
- IPC0ISET0 register
- IPC0TXD0 register
- IPC0RXD0 register
- IPC0CLR0 register
- IPCPAR.PAIPCIR0 bit

SAIPCIR1 bit

Security attributes of registers. The target registers are as follow.

- IPC0STA1 register
- IPC0ISET1 register
- IPC0TXD1 register
- IPC0RXD1 register
- IPC0CLR1 register
- IPCPAR.PAIPCIR1 bit

SAIPCIR2 bit

Security attributes of registers. The target registers are as follow.

- IPC1STA0 register
- IPC1ISET0 register
- IPC1TXD0 register
- IPC1RXD0 register
- IPC1CLR0 register
- IPCPAR.PAIPCIR2 bit

SAIPCIR3 bit

Security attributes of registers. The target registers are as follow.

- IPC1STA1 register
- IPC1ISET1 register
- IPC1TXD1 register
- IPC1RXD1 register
- IPC1CLR1 register
- IPCPAR.PAIPCIR3 bit

3.2.2 IPCPAR : IPC Privileged Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x614

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PAIPC IR3	PAIPC IR2	PAIPC IR1	PAIPC IR0
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PAIPC NMI1	PAIPC NMI0	—	—	—	—	—	—	PAIPC SEM1	PAIPC SEM0
Value after reset:	x	x	x	x	x	x	1	1	x	x	x	x	x	x	1	1

Bit	Symbol	Function	R/W
0	PAIPCSEM0	Privileged attributes of registers for IPCSEMn (n = 0 to 7) 0: Privileged 1: Unprivileged	R/W
1	PAIPCSEM1	Privileged attributes of registers for IPCSEMn (n = 8 to 15) 0: Privileged 1: Unprivileged	R/W
7:2	—	The read values are undefined. The write value should be 1.	R/W
8	PAIPC NMI0	Privileged attributes of registers for IPC0NMISTA, IPC0NMISSET and IPC0NMICLR 0: Privileged 1: Unprivileged	R/W
9	PAIPC NMI1	Privileged attributes of registers for IPC1NMISTA, IPC1NMISSET and IPC1NMICLR 0: Privileged 1: Unprivileged	R/W
15:10	—	The read values are undefined. The write value should be 1.	R/W
16	PAIPCIR0	Privileged attributes of registers for IPC0STA0, IPC0ISET0, IPC0TXD0, IPC0RXD0 and IPC0CLR0 0: Privileged 1: Unprivileged	R/W
17	PAIPCIR1	Privileged attributes of registers for IPC0STA1, IPC0ISET1, IPC0TXD1, IPC0RXD1 and IPC0CLR1 0: Privileged 1: Unprivileged	R/W
18	PAIPCIR2	Privileged attributes of registers for IPC1STA0, IPC1ISET0, IPC1TXD0, IPC1RXD0 and IPC1CLR0 0: Privileged 1: Unprivileged	R/W
19	PAIPCIR3	Privileged attributes of registers for IPC1STA1, IPC1ISET1, IPC1TXD1, IPC1RXD1 and IPC1CLR1 0: Privileged 1: Unprivileged	R/W
31:20	—	The read values are undefined. The write value should be 1.	R/W

Note: S-TYPE-2, P-TYPE-1

The Security Attribution of bits of this register is determined by the corresponding bits of IPCSAR.

This register is the write protected bit by bit according to the security attribute setting.

Secure bits is write-protected by PRCCR_S.PRC4.

Non-secure bits is write-protected by PRCCR_NS.PRC4.

PAIPCSEM0 bit

Privileged attributes of registers. The target registers are as follow.

- IPCSEM0 to IPCSEM7 registers

PAIPCSEM1 bit

Privileged attributes of registers. The target registers are as follow.

- IPCSEM8 to IPCSEM15 registers

PAIPCNMI0 bit

Privileged attributes of registers. The target registers are as follow.

- IPC0NMISTA register
- IPC0NMISSET register
- IPC0NMICLR register

PAIPCNMI1 bit

Privileged attributes of registers. The target registers are as follow.

- IPC1NMISTA register
- IPC1NMISSET register
- IPC1NMICLR register

PAIPCIR0 bit

Privileged attributes of registers. The target registers are as follow.

- IPC0STA0 register
- IPC0ISET0 register
- IPC0TXD0 register
- IPC0RXD0 register
- IPC0CLR0 register

PAIPCIR1 bit

Privileged attributes of registers. The target registers are as follow.

- IPC0STA1 register
- IPC0ISET1 register
- IPC0TXD1 register
- IPC0RXD1 register
- IPC0CLR1 register

PAIPCIR2 bit

Privileged attributes of registers. The target registers are as follow.

- IPC1STA0 register
- IPC1ISET0 register
- IPC1TXD0 register
- IPC1RXD0 register
- IPC1CLR0 register

PAIPCIR3 bit

Privileged attributes of registers. The target registers are as follow.

- IPC1STA1 register
- IPC1ISET1 register
- IPC1TXD1 register

- IPC1RXD1 register
- IPC1CLR1 register

3.2.3 IPCSEMn : Semaphore Register n (n = 0 to 15)

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x000 + 0x4 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LOCK
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LOCK	This bit indicates the shared resource is locked. Set condition : Reading this register. Clear condition : Writing 1 to this bit. 0: Unlocked 1: Locked	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: The LOCK bit is not set for reads with halfword access or byte access that do not include the b0.

Note: S-TYPE-3, P-TYPE-3

The IPC has the registers to perform semaphore for synchronization and exclusive control between CPU0 and CPU1. This register only indicates the status and does not have any hardware protection of shared memory. Thus, the exclusive control must be done by software that uses this register.

The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

LOCK bit (Shared resource is locked for Semaphore Control)

The LOCK bit of this register is used to achieve mutually exclusive access between processors. By using this bit, the application developer can select whether to use the hardware semaphore separately or in common between the two processors.

3.2.4 IPC0NMISTA : Inter-Processor0 NMI Request Status Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NMI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NMI	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
31:1	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

The
IPC0NMISTA

register indicates that a non-maskable interrupt is issued from CPU0 to CPU1. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

NMI bit (NMI Request Set)

The NMI bit indicates an NMI interrupt request is issued from CPU0 to CPU1.

3.2.5 IPC0NMISSET : Inter-Processor0 NMI Request Set Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x084

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SET
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SET	Writing 1 to the bit sets IPC0NMISTA.NMI.	W
31:1	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC0NMISSET

register is used to issue a non-maskable interrupt from CPU0 to CPU1. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SET bit (NMI Request Set)

When 1 is written, IPC0NMISTA.NMI is set to 1 and an NMI interrupt is issued to CPU0.

3.2.6 IPC0NMICLR : Inter-Processor0 NMI Request Clear Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x088

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLR	Writing 1 to this bit clears IPC0NMISTA.NMI.	W
31:1	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC0NMICLR

register is used for canceling non-maskable interrupt from CPU1 to CPU0. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLR bit (NMI Request Clear)

When 1 is written, IPC0NMISTA.NMI is clear to 0 and the NMI interrupt to CPU0 is canceled.

3.2.7 IPC1NMISTA : Inter-Processor1 NMI Request Status Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NMI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NMI	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
31:1	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

The
IPC1NMISTA

register indicates that a non-maskable interrupt is issued from CPU0 to CPU1. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

NMI bit (NMI Request Status)

The NMI bit indicates an NMI interrupt request is issued from CPU0 to CPU1.

3.2.8 IPC1NMISSET : Inter-Processor1 NMI Request Set Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x094

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SET
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SET	Writing 1 to this bit sets IPC1NMISTA.NMI.	W
31:1	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC1NMISSET

register is used to issue a non-maskable interrupt from CPU0 to CPU1. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SET bit (NMI Request Set)

When 1 is written, IPC1NMISTA.NMI is set to 1 and an NMI interrupt is issued to CPU1.

3.2.9 IPC1NMICLR : Inter-Processor1 NMI Request Clear Register

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x098

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLR	Writing 1 to this bit clears IPC1NMISTA.NMI.	W
31:1	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC1NMICLR

register is used to cancel non-maskable interrupt from CPU0 to CPU1. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLR bit (NMI Request Clear)

When 1 is written, IPC1NMISTA.NMI is clear to 0 and the NMI interrupt to CPU1 is canceled.

3.2.10 IPC0STA0 : Inter-Processor0 Status Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FERR	RERR	—	—	—	—	—	—	FULL	RDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	IRQ7	IRQ6	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	IRQ7 to IRQ0	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
15:8	—	These bits are read as 0. The write value should be 0.	R
16	RDY	This bit is set when FIFO is not empty. This bit is used for the interrupt request. 0: FIFO 00 data is not ready 1: FIFO 00 data is ready	R
17	FULL	0: FIFO 00 is not full 1: FIFO 00 is full	R
23:18	—	These bits are read as 0. The write value should be 0.	R
24	RERR	Indicates that the message FIFO 00 tried to read data despite being empty. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
25	FERR	Indicates that the message FIFO 00 tried to send more data even though it was full. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
31:26	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

This register indicates the status of the maskable interrupt IPC0IRQ0 from CPU1 to CPU0. When any of IRQn (n = 0 to 7), RDY, RERR, or FERR is 1, the IPC0IRQ0 interrupt is issued. Read this register in the interrupt handler to determine which interrupt has occurred. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

IRQn bit (Interrupt Request Status n (n = 0 to 7))

These bits indicate an interrupt to CPU0. Writing "1" to IPC0ISET0.SETn sets IRQn to "1". Writing "1" to IPC0CLR0.CLRn clears IRQn to "0". If any bit is "1", an IPC0IRQ0 interrupt request is issued.

RDY bit (Receive Data Ready)

When this bit is "1", it indicates that there is data received in the message FIFO 00 for CPU0. When the message FIFO is empty, this bit is "0". When this bit is "1", the IPC0IRQ0 interrupt request is issued.

FULL bit (FIFO Full)

The depth of message FIFO 00 is 4. When this bit is "1", it indicates that the message FIFO 00 is full.

RERR bit (RDY Error Status)

When reading IPC0RXD0 while the RDY bit is "0", this bit sets to "1". Writing "1" to the IPC0CLR0.RCLR bit clears this bit to "0". When this bit is "1", the IPC0IRQ0 interrupt request is issued.

FERR bit (FULL Error Status)

When writing to IPC0TXD0 while the FULL bit is "1", this bit sets to "1". Writing "1" to the IPC0CLR0.FCLR bit clears this bit to "0". When this bit is "1", the IPC0IRQ0 interrupt request is issued.

3.2.11 IPC0SET0 : Inter-Processor0 IRQ Request Set Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SET7	SET6	SET5	SET4	SET3	SET2	SET1	SET0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SETn (n = 0 to 7)	Writing 1 to this bit sets IPC0STA0.IRQn	W
31:8	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The

IPC0SET0

register is used to issue a maskable interrupt from CPU0 to CPU1. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SETn bit (Interrupt Request Set n (n = 0 to 7))

When 1 is written, IPC0STA0.IRQn is set to 1 and a maskable interrupt to CPU0 is issued.

3.2.12 IPC0TXD0 : Inter-Processor0 FIFO Transfer Data Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0C8

Bit position:	31	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
Bit field:	TXD[31:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	TXD[31:0]	Transmit data for the Message FIFO 00	W

Note: Only writing with a 32-bit data size is valid. Halfword or byte access is ignored.

Note: S-TYPE-3, P-TYPE-3

This is a register for writing data to message FIFO 00 from CPU1 to CPU0. Write to this register by CPU1. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

TXD[31:0] bits (Transmit data for the Message FIFO 00)

Writing to this register sets the transmit data in message FIFO 00 and the IPC0STA0.RDY bit becomes "1". If you write to this register while the IPC0STA0.FULL bit is "1", the write operation is ignored, and the IPC0STA0.FERR bit becomes "1".

3.2.13 IPC0RXD0 : Inter-Processor0 FIFO Receive Data Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0CC

Bit position: 31

0

Bit field:

RXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RXD[31:0]	Received data from the message FIFO 00 Reading this register updates it with the next received data in the message FIFO 00.	R

Note: If a TRZ error occurs, the read is 0 and it does not update to the next data.
S-TYPE-3, P-TYPE-3

This register is for reading received data from the message FIFO 00 from CPU1 to CPU0. Read this register by CPU0. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

RXD[31:0] bits (Received data from the message FIFO 00)

If this register is read while the IPC0STA0.RDY bit is "0", the IPC0STA0.RERR bit is set to "1".

3.2.14 IPC0CLR0 : Inter-Processor0 Clear Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0D0

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	FCLR	RCLR	—	—	—	—	—	—	—	RST
---	---	---	---	---	---	------	------	---	---	---	---	---	---	---	-----

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	—	CLR7	CLR6	CLR5	CLR4	CLR3	CLR2	CLR1	CLR0
---	---	---	---	---	---	---	---	------	------	------	------	------	------	------	------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	CLRn (n = 0 to 7)	Writing 1 to this bit clears IPC0STA0.IRQn.	W
15:8	—	These bits are read as 0. The write value should be 0.	W
16	RST	Writing 1 resets Message FIFO 00.	W
23:17	—	These bits are read as 0. The write value should be 0.	W
24	RCLR	Writing 1 resets IPC0STA0.RERR.	W
25	FCLR	Writing 1 resets IPC0STA0.FERR.	W
31:26	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The

IPC0CLR0

register is used to cancel the maskable interrupt from CPU1 to CPU0. The register can then be used to cancel the FIFO transfer. It can also clear the Error flag that occurred in the FIFO. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLRn bits (Interrupt Request Clear n (n = 0 to 7))

When 1 is written, IPC0STA0.IRQn is clear to 0. All bits of IPC0STA0 are 0 and maskable interrupts to CPU0 are cancelled.

RST bit (FIFO Reset)

When 1 is written, the RST bit resets Message FIFO 00. The data stored in the FIFO becomes invalid and the following registers are cleared:

- IPC0STA0.FULL
- IPC0STA0.RDY

If there is no interrupt event other than the message FIFO, the interrupt request is canceled.

RCLR bit (Clear RDY Error Status 00)

The RCLR bit clears IPC0STA0.RERR.

FCLR bit (Clear FULL Error Status 00)

The FCLR bit clears IPC0STA0.FERR.

3.2.15 IPC0STA1 : Inter-Processor0 Status Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0E0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FERR	RERR	—	—	—	—	—	—	FULL	RDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	IRQ7	IRQ6	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	IRQn (n = 0 to 7)	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
15:8	—	These bits are read as 0. The write value should be 0.	R
16	RDY	This bit is set when FIFO is not empty. This bit is used for the interrupt request. 0: FIFO 01 data is not ready 1: FIFO 01 data is ready	R
17	FULL	0: FIFO 01 is not full 1: FIFO 01 is full	R
23:18	—	These bits are read as 0. The write value should be 0.	R
24	RERR	Indicates that the message FIFO 01 tried to read data despite being empty. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R

Bit	Symbol	Function	R/W
25	FERR	Indicates that the message FIFO 01 tried to send more data even though it was full. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
31:26	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

This register indicates the status of the maskable interrupt IPC0IRQ1 from CPU1 to CPU0. When any of IRQn (n = 0 to 7), RDY, RERR, or FERR is '1', the IPC0IRQ1 interrupt is issued. Read this register in the interrupt handler to determine which interrupt has occurred. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

IRQn bit (Interrupt Request Status n (n = 0 to 7))

These bits indicate an interrupt to CPU0. Writing "1" to IPC0ISET1.SETn sets IRQn to "1". Writing "1" to IPC0CLR1.CLRn clears IRQn to "0". If any bit is "1", an IPC0IRQ1 interrupt request is issued.

RDY bit (Receive Data Ready)

When this bit is "1", it indicates that there is data received in the message FIFO 01 for CPU0. When the message FIFO is empty, this bit is "0". When this bit is "1", the IPC0IRQ1 interrupt request is issued.

FULL bit (FIFO Full)

The depth of message FIFO 01 is 4. When this bit is "1", it indicates that the message FIFO 01 is full.

RERR bit (RDY Error Status)

When reading IPC0RXD1 while the RDY bit is "0", this bit sets to "1". Writing "1" to the IPC0CLR1.RCLR bit clears this bit to "0". When this bit is "1", the IPC0IRQ1 interrupt request is issued.

FERR bit (FULL Error Status)

When writing to IPC0TXD1 while the FULL bit is "1", this bit sets to "1". Writing "1" to the IPC0CLR1.FCLR bit clears this bit to "0". When this bit is "1", the IPC0IRQ1 interrupt request is issued.

3.2.16 IPC0ISET1 : Inter-Processor0 IRQ Request Set Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0E4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SET7	SET6	SET5	SET4	SET3	SET2	SET1	SET0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SETn (n = 0 to 7)	Writing 1 to this bit sets IPC0STA1.IRQn.	W
31:8	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC0ISET1

register is used to issue a maskable interrupt from CPU1 to CPU0. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SETn bit (Interrupt Request Set n (n = 0 to 7))

When 1 is written, IPC0STA1.IRQn is set to 1 and a maskable interrupt to CPU0 is issued.

3.2.17 IPC0TXD1 : Inter-Processor0 FIFO Transfer Data Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0E8

Bit position: 31

0

Bit field:

TXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TXD[31:0]	Transmit data for the Message FIFO 01	W

Note: Only writing with a 32-bit data size is valid. Halfword or byte access is ignored.

Note: S-TYPE-3, P-TYPE-3

This is a register for writing data to message FIFO 01 from CPU1 to CPU0. Write to this register by CPU1. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

TXD[31:0] bits (Transmit data for the Message FIFO 01)

Writing to this register sets the transmit data in message FIFO 01 and the IPC0STA1.RDY bit becomes "1". If you write to this register while the IPC0STA1.FULL bit is "1", the write operation is ignored, and the IPC0STA1.FERR bit becomes "1".

3.2.18 IPC0RXD1 : Inter-Processor0 FIFO Receive Data Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0EC

Bit position: 31

0

Bit field:

RXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RXD[31:0]	Received data from the message FIFO 01 Reading this register updates it with the next received data in the message FIFO 01.	R

Note: If a TRZ error occurs, the read is 0 and it does not update to the next data.

S-TYPE-3, P-TYPE-3

This register is for reading received data from the message FIFO 01 from CPU1 to CPU0. Read this register by CPU0. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

RXD[31:0] bits (Received data from the message FIFO 01)

If this register is read while the IPC0STA1.RDY bit is "0", the IPC0STA1.RERR bit is set to "1".

3.2.19 IPC0CLR1 : Inter-Processor0 Clear Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x0F0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FCLR	RCLR	—	—	—	—	—	—	—	RST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CLR7	CLR6	CLR5	CLR4	CLR3	CLR2	CLR1	CLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CLRn (n = 0 to 7)	Writing 1 to this bit clears IPC0STA1.IRQn.	W
15:8	—	These bits are read as 0. The write value should be 0.	W
16	RST	Writing 1, then resets Message FIFO 01.	W
23:17	—	These bits are read as 0. The write value should be 0.	W
24	RCLR	Writing 1 resets IPC0STA1.RERR.	W
25	FCLR	Writing 1 resets IPC0STA1. FERR.	W
31:26	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC0CLR1

register is used to cancel the maskable interrupt from CPU1 to CPU0. The register can then be used to cancel the FIFO transfer. This register can also clear the Error flag that occurred in the FIFO. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLRn bit (Interrupt Request Clear n (n = 0 to 7))

When 1 is written, IPC0STA1.IRQn is cleared to 0. All bits of IPC0STA1 are 0, and maskable interrupts to CPU0 are cancelled.

RST bit (FIFO Reset)

Writing 1 to the RST bit resets message FIFO 01. The data stored in the FIFO becomes invalid and the following registers are cleared:

- IPC0STA1.FULL
- IPC0STA1.RDY

If there is no interrupt event other than the message FIFO, the interrupt request is canceled.

RCLR bit (Clear RDY Error Status 01)

The RCLR bit clears IPC0STA1.RERR.

FCLR bit (Clear FULL Error Status 01)

The FCLR bit clears IPC0STA1. FERR.

3.2.20 IPC1STA0 : Inter-Processor1 Status Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FERR	RERR	—	—	—	—	—	—	FULL	RDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	IRQ7	IRQ6	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	IRQn (n = 0 to 7)	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
15:8	—	These bits are read as 0. The write value should be 0.	R
16	RDY	This bit is set when FIFO is not empty. This bit is used for the interrupt request. 0: FIFO 10 data is not ready 1: FIFO 10 data is ready	R
17	FULL	0: FIFO 10 is not full 1: FIFO 10 is full	R
23:18	—	These bits are read as 0. The write value should be 0.	R
24	RERR	This bit indicates that the message FIFO 10 tried to read data despite being empty. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
25	FERR	This bit indicates that the message FIFO 10 tried to send more data even though it was full. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
31:26	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

This register indicates the status of the maskable interrupt IPC1IRQ0 from CPU0 to CPU1. When any of IRQn (n = 0 to 7), RDY, RERR, or FERR is '1', the IPC1IRQ0 interrupt is issued. Read this register in the interrupt handler to determine which interrupt has occurred. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

IRQn bit (Interrupt Request Status n (n = 0 to 7))

These bits indicate an interrupt to CPU1. Writing "1" to IPC1ISET0.SETn sets IRQn to "1". Writing "1" to IPC1CLR0.CLRn clears IRQn to "0". If any bit is "1", an IPC1IRQ0 interrupt request is issued.

RDY bit (Receive Data Ready)

When this bit is "1", it indicates that there is data received in the message FIFO 10 for CPU1. When the message FIFO is empty, this bit is "0". When this bit is "1", the IPC1IRQ0 interrupt request is issued.

FULL bit (FIFO Full)

The depth of message FIFO 10 is 4. When this bit is "1", it indicates that the message FIFO 10 is full.

RERR bit (RDY Error Status)

When reading IPC1RXD0 while the RDY bit is "0", this bit sets to "1". Writing "1" to the IPC1CLR0.RCLR bit clears this bit to "0". When this bit is "1", the IPC1IRQ0 interrupt request is issued.

FERR bit (FULL Error Status)

When writing to IPC1TXD0 while the FULL bit is "1", this bit sets to "1". Writing "1" to the IPC1CLR0.FCLR bit clears this bit to "0". When this bit is "1", the IPC1IRQ0 interrupt request is issued.

3.2.21 IPC1ISET0 : Inter-Processor1 IRQ Request Set Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SET7	SET6	SET5	SET4	SET3	SET2	SET1	SET0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SETn (n = 0 to 7)	Writing 1 to this bit sets IPC1STA0.IRQn.	W
31:8	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The

IPC1ISET0

register is used to issue a maskable interrupt from CPU0 to CPU1. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SETn bit (Interrupt Request Set n (n = 0 to 7))

When 1 is written, IPC1STA0.IRQn is set to 1 and a maskable interrupt to CPU1 is issued.

3.2.22 IPC1TXD0 : Inter-Processor1 FIFO Transfer Data Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x108

Bit position:	31	0																																		
Bit field:	TXD[31:0]																																			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	TXD[31:0]	Transmit data for the Message FIFO 10	W

Note: Only writing with a 32-bit data size is valid. Halfword or byte access is ignored.

Note: S-TYPE-3, P-TYPE-3

This is a register for writing data to message FIFO 10 from CPU0 to CPU1. Write to this register by CPU0. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

TXD[31:0] bits (Transmit data for the Message FIFO 10)

Writing to this register sets the transmit data in message FIFO 10 and the IPC1STA0.RDY bit becomes "1". If you write to this register while the IPC1STA0.FULL bit is "1", the write operation is ignored, and the IPC1STA0.FERR bit becomes "1".

3.2.23 IPC1RXD0 : Inter-Processor1 FIFO Receive Data Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x10C

Bit position: 31

0

Bit field:

RXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RXD[31:0]	Received data from the message FIFO 10 Reading this register updates it with the next received data in the message FIFO 10.	R

Note: If a TRZ error occurs, the read is 0 and it does not update to the next data.
S-TYPE-3, P-TYPE-3

This register is for reading received data from the message FIFO 10 from CPU0 to CPU1. Read this register by CPU1. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

RXD[31:0] bits (Received data from the message FIFO 10)

If this register is read while the IPC1STA0.RDY bit is "0", the IPC1STA0.RERR bit is set to "1".

3.2.24 IPC1CLR0 : Inter-Processor1 Request Clear Register0

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x110

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	FCLR	RCLR	—	—	—	—	—	—	—	RST
---	---	---	---	---	---	------	------	---	---	---	---	---	---	---	-----

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	—	CLR7	CLR6	CLR5	CLR4	CLR3	CLR2	CLR1	CLR0
---	---	---	---	---	---	---	---	------	------	------	------	------	------	------	------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	CLRn (n = 0 to 7)	Writing 1 to this bit clears IPC1STA0.IRQn.	W
15:8	—	These bits are read as 0. The write value should be 0.	W
16	RST	Writing 1 resets Message FIFO 10.	W
23:17	—	These bits are read as 0. The write value should be 0.	W
24	RCLR	Writing 1 resets IPC1STA0.RERR.	W
25	FCLR	Writing 1 resets IPC1STA0. FERR.	W
31:26	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC1CLR0

register is used to cancel the maskable interrupt from CPU0 to CPU1. The register can then be used to cancel the FIFO transfer. It can also clear the Error flag that occurred in the FIFO. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLRn bit (Interrupt Request Clear n (n = 0 to 7))

When 1 is written, IPC1STA0.IRQn is cleared to 0. All bits of IPC1STA0 are 0, and maskable interrupts to CPU1 are cancelled.

RST bit (FIFO Reset)

Writing 1 to the RST bit resets message FIFO 10. The data stored in the FIFO becomes invalid and the following registers are cleared:

- IPC1STA0.FULL
- IPC1STA0.RDY

If there is no interrupt event other than the message FIFO, the interrupt request is canceled.

RCLR bit (Clear RDY Error Status 10)

The RCLR bit clears IPC1STA0.RERR.

FCLR bit (Clear FULL Error Status 10)

The FCLR bit clears IPC1STA0.FERR.

3.2.25 IPC1STA1 : Inter-Processor1 Request Status Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x120

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FERR	RERR	—	—	—	—	—	—	FULL	RDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	IRQ7	IRQ6	IRQ5	IRQ4	IRQ3	IRQ2	IRQ1	IRQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	IRQn (n = 0 to 7)	This bit indicates the status of interrupt request. 0: Not requesting interrupt 1: Requesting interrupt	R
15:8	—	These bits are read as 0. The write value should be 0.	R
16	RDY	This bit is set when FIFO is not empty. This bit is used for the interrupt request. 0: FIFO 11 data is not ready 1: FIFO 11 data is ready	R
17	FULL	0: FIFO 11 is not full 1: FIFO 11 is full	R
23:18	—	These bits are read as 0. The write value should be 0.	R
24	RERR	Indicates that the message FIFO 11 tried to read data despite being empty. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
25	FERR	Indicates that the message FIFO 11 tried to send more data even though it was full. 0: Not Error. Not requesting interrupt. 1: Error. Requesting interrupt.	R
31:26	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

This register indicates the status of the maskable interrupt IPC1IRQ1 from CPU0 to CPU1. When any of IRQn (n = 0 to 7), RDY, RERR, or FERR is '1', the IPC1IRQ1 interrupt is issued. Read this register in the interrupt handler to determine which interrupt has occurred. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

IRQn bit (Interrupt Request Status n (n = 0 to 7))

These bits indicate an interrupt to CPU1. Writing "1" to IPC1ISET1.SETn sets IRQn to "1". Writing "1" to IPC1CLR1.CLRn clears IRQn to "0". If any bit is "1", an IPC1IRQ1 interrupt request is issued.

RDY bit (Receive Data Ready)

When this bit is "1", it indicates that there is data received in the message FIFO 11 for CPU1. When the message FIFO is empty, this bit is "0". When this bit is "1", the IPC1IRQ1 interrupt request is issued.

FULL bit (FIFO Full)

The depth of message FIFO 11 is 4. When this bit is "1", it indicates that the message FIFO 11 is full.

RERR bit (RDY Error Status)

When reading IPC1RXD1 while the RDY bit is "0", this bit sets to "1". Writing "1" to the IPC1CLR1.RCLR bit clears this bit to "0". When this bit is "1", the IPC1IRQ1 interrupt request is issued.

FERR bit (FULL Error Status)

When writing to IPC1TXD1 while the FULL bit is "1", this bit sets to "1". Writing "1" to the IPC1CLR1.FCLR bit clears this bit to "0". When this bit is "1", the IPC1IRQ1 interrupt request is issued.

3.2.26 IPC1ISET1 : Inter-Processor1 IRQ Request Set Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x124

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SET7	SET6	SET5	SET4	SET3	SET2	SET1	SET0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SETn (n = 0 to 7)	Writing 1 to this bit sets IPC1STA1.IRQn.	W
31:8	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The

IPC1ISET1

register is used to issue a maskable interrupt from CPU0 to CPU1. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

SETn bit (Interrupt Request Set n (n = 0 to 7))

When 1 is written, IPC1STA1.IRQn is set to 1 and a maskable interrupt to CPU1 is issued.

3.2.27 IPC1TXD1 : Inter-Processor1 FIFO Transfer Data Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x128

Bit position: 31

0

Bit field:

TXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TXD[31:0]	Transmit data for the Message FIFO 11	W

Note: Only writing with a 32-bit data size is valid. Halfword or byte access is ignored.

Note: S-TYPE-3, P-TYPE-3

This is a register for writing data to message FIFO 11 from CPU0 to CPU1. Write to this register by CPU0. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

TXD[31:0] bits (Transmit data for the Message FIFO 11)

Writing to this register sets the transmit data in message FIFO 11 and the IPC1STA1.RDY bit becomes "1". If you write to this register while the IPC1STA1.FULL bit is "1", the write operation is ignored, and the IPC1STA1.FERR bit becomes "1".

3.2.28 IPC1RXD1 : Inter-Processor1 FIFO Receive Data Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x12C

Bit position: 31

0

Bit field:

RXD[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RXD[31:0]	Received data from the message FIFO 11 Reading this register updates it with the next received data in the message FIFO 11.	R

Note: If a TRZ error occurs, the read is 0 and it does not update to the next data.

S-TYPE-3, P-TYPE-3

This register is for reading received data from the message FIFO 11 from CPU0 to CPU1. Read this register by CPU1. The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR.

RXD[31:0] bits (Received data from the message FIFO 11)

If this register is read while the IPC1STA1.RDY bit is "0", the IPC1STA1.RERR bit is set to "1".

3.2.29 IPC1CLR1 : Inter-Processor1 Clear Register1

Base address: IPC = 0x4002_0000
IPC_NS = 0x5002_0000

Offset address: 0x130

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	FCLR	RCLR	—	—	—	—	—	—	—	RST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CLR7	CLR6	CLR5	CLR4	CLR3	CLR2	CLR1	CLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CLRn (n = 0 to 7)	Writing 1 to this bit clears IPC0IRQSTA1.IRQn.	W
15:8	—	These bits are read as 0. The write value should be 0.	W
16	RST	Writing 1 resets message FIFO 11.	W
23:17	—	These bits are read as 0. The write value should be 0.	W
24	RCLR	Writing 1 resets IPC1STA1.RERR.	W
25	FCLR	Writing 1 resets IPC1STA1.FERR.	W
31:26	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The
IPC1CLR1

register is used to cancel the maskable interrupt from CPU0 to CPU1. The register can then be used to cancel the FIFO transfer. It can also clear the Error flag that occurred in the FIFO. The IRQ is assigned eight events, but there is only one interrupt event. The security of this register depends on IPCSAR. The privilege of this register depends on IPCPAR.

CLRn bit (Interrupt Request Clear n (n = 0 to 7))

When 1 is written, IPC1STA1.IRQn is cleared to 0. All bits of IPC1STA1 are 0, and maskable interrupts to CPU1 are cancelled.

RST bit (FIFO Reset)

Writing 1 resets the message FIFO 11. The data stored in the FIFO becomes invalid and the following registers are cleared:

- IPC1STA1.FULL
- IPC1STA1.RDY

If there is no interrupt event other than the message FIFO, the interrupt request is canceled.

RCLR bit (Clear RDY Error Status 11)

The RCLR bit clears IPC1STA1.RERR.

FCLR bit (Clear FULL Error Status 11)

The FCLR bit clears IPC1STA1.FERR.

3.3 Operation

3.3.1 Semaphore

The IPC has registers to perform semaphore for synchronization and exclusive control between CPU0 and CPU1.

Figure 3.2 shows the structure of the IPCSEM register used for semaphore control.

This register only indicates the status and does not have any hardware protection of shared memory. Thus, the exclusive control must be done by software that uses this register.

Security attributes and privileged attributes must be set in common for IPCSEM0 to IPCSEM7, and IPCSEM8 to IPCSEM15.

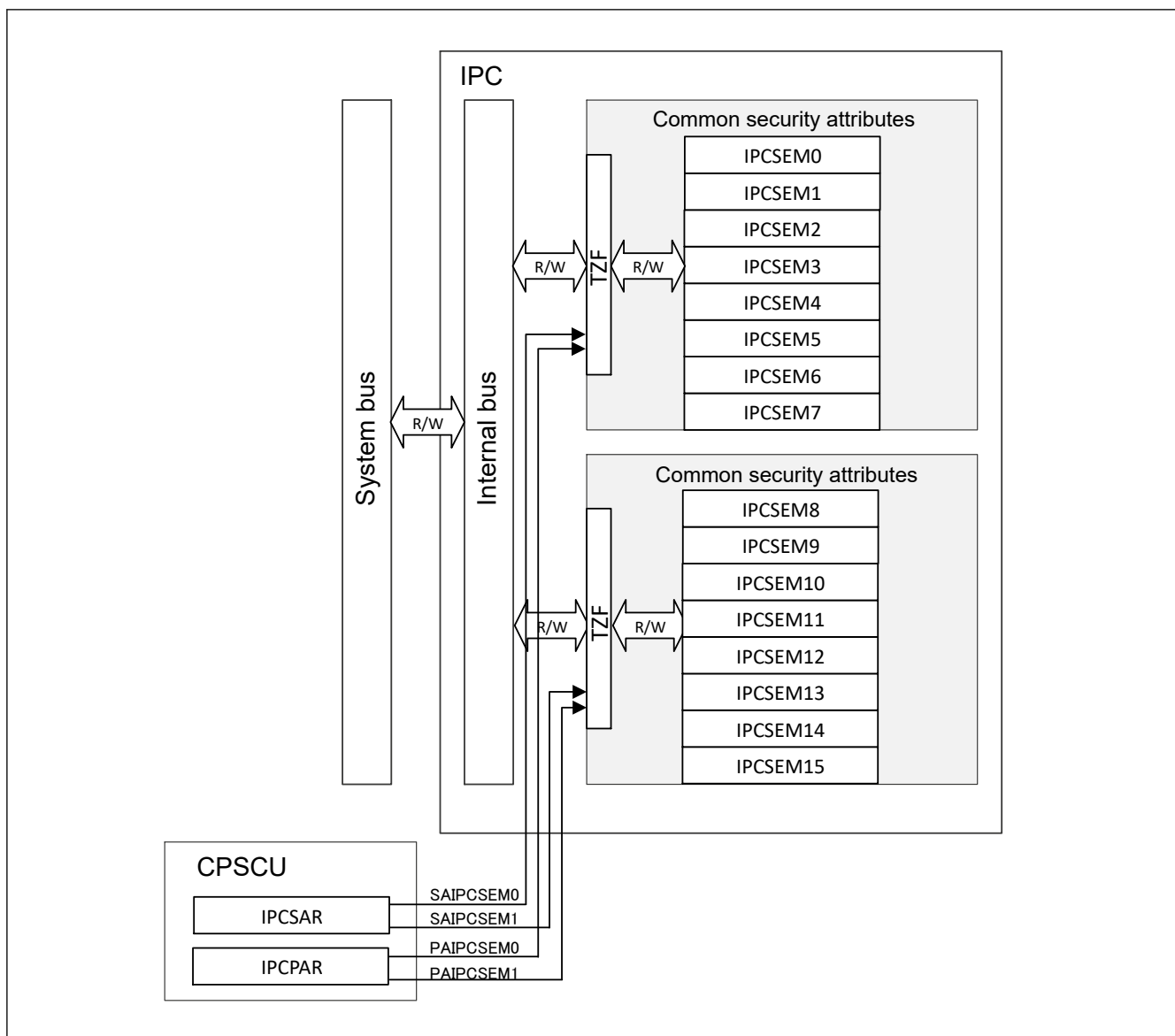


Figure 3.2 IPCSEM register

3.3.2 Inter-processor Non-maskable Interrupt

IPC supports the Inter-processor Non-maskable Interrupt feature between CPU0 and CPU1. Figure 3.3 shows the configuration diagram of the "Inter-processor Non-maskable Interrupt" function. IPC0NMI is the non-maskable interrupt from CPU1 to CPU0 and IPC1NMI is the one from CPU0 to CPU1. IPCiNMI ($i = 0, 1$) functions are controlled by IPCiNMISTA, IPCiNMISSET, and IPCiNMICLR. The security attributes of these bits are controlled by IPCSAR and privileged attributes of these bits are controlled by IPCPAR.

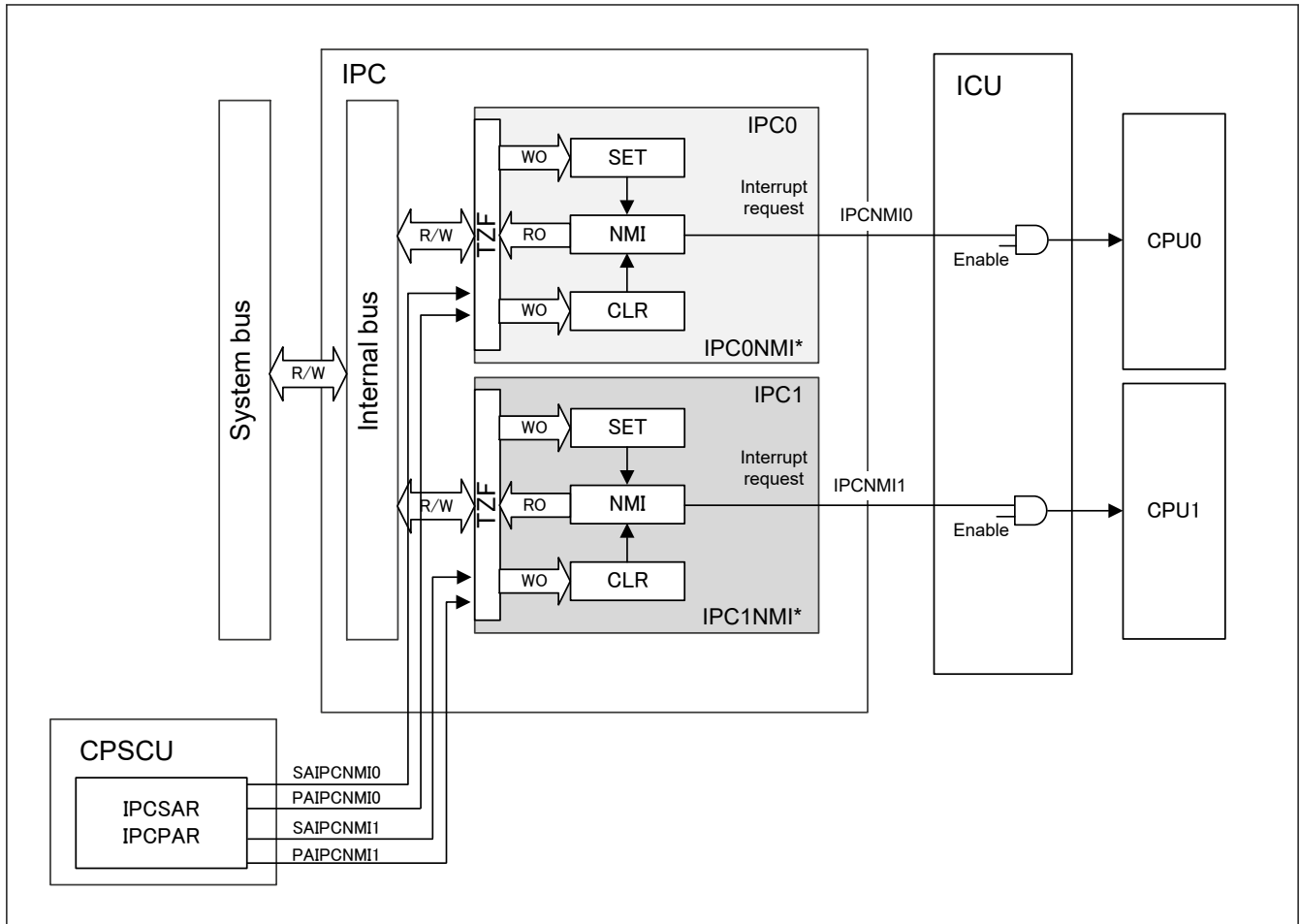


Figure 3.3 Inter-processor non-maskable interrupt

The CPU can only select either Secure or Non-secure for NMI. The IPC NMI security attribute must also be set accordingly. See [section 14, Interrupt Controller Unit \(ICU\)](#) for controls on mixing Secure and Non-secure NMIs. If the IPC does not use NMI interrupt, mask it in the ICU register.

3.3.3 Inter-processor Maskable Interrupt

IPC supports the Inter-processor Maskable Interrupt feature between CPU0 and CPU1. [Figure 3.4](#) shows the configuration diagram of the "Inter-processor Maskable Interrupt" function. IPC0IRQj (j = 0, 1) is the maskable interrupt from CPU1 to CPU0 and IPC1IRQj (j = 0, 1) is the maskable interrupt from CPU0 to CPU1. IPCiIRQj (i = 0, 1), (j = 0, 1) are controlled by IPCiSTAj.IRQn, IPCiSETj.SETn, IPCiCLRj.CLRn (n = 0 to 7) respectively. The security attributes of the registers related to this function are controlled by IPCSAR, and the privileged attributes are controlled by IPCPAR. Security attributes and privileged attributes can be set for each IPCiIRQj (i = 0, 1), (j = 0, 1). Each IPCiIRQj has eight interrupt request sources, IPCiSTAj.IRQn (n = 0 to 7).

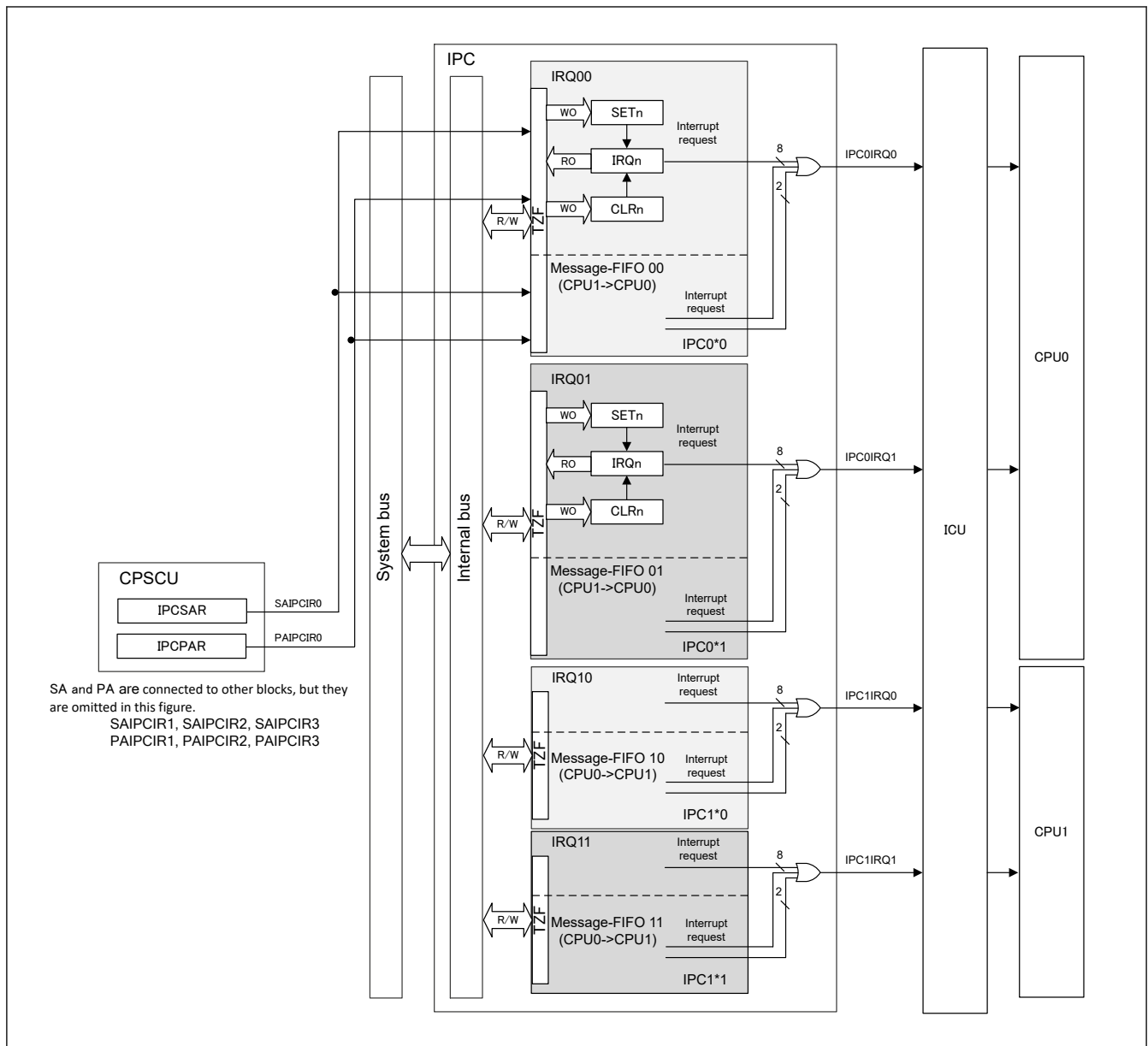


Figure 3.4 Inter-processor maskable interrupt

3.3.4 Exclusive Control Flow

Figure 3.5 shows an example of the exclusive access control flow when semaphore is executed between two CPUs.

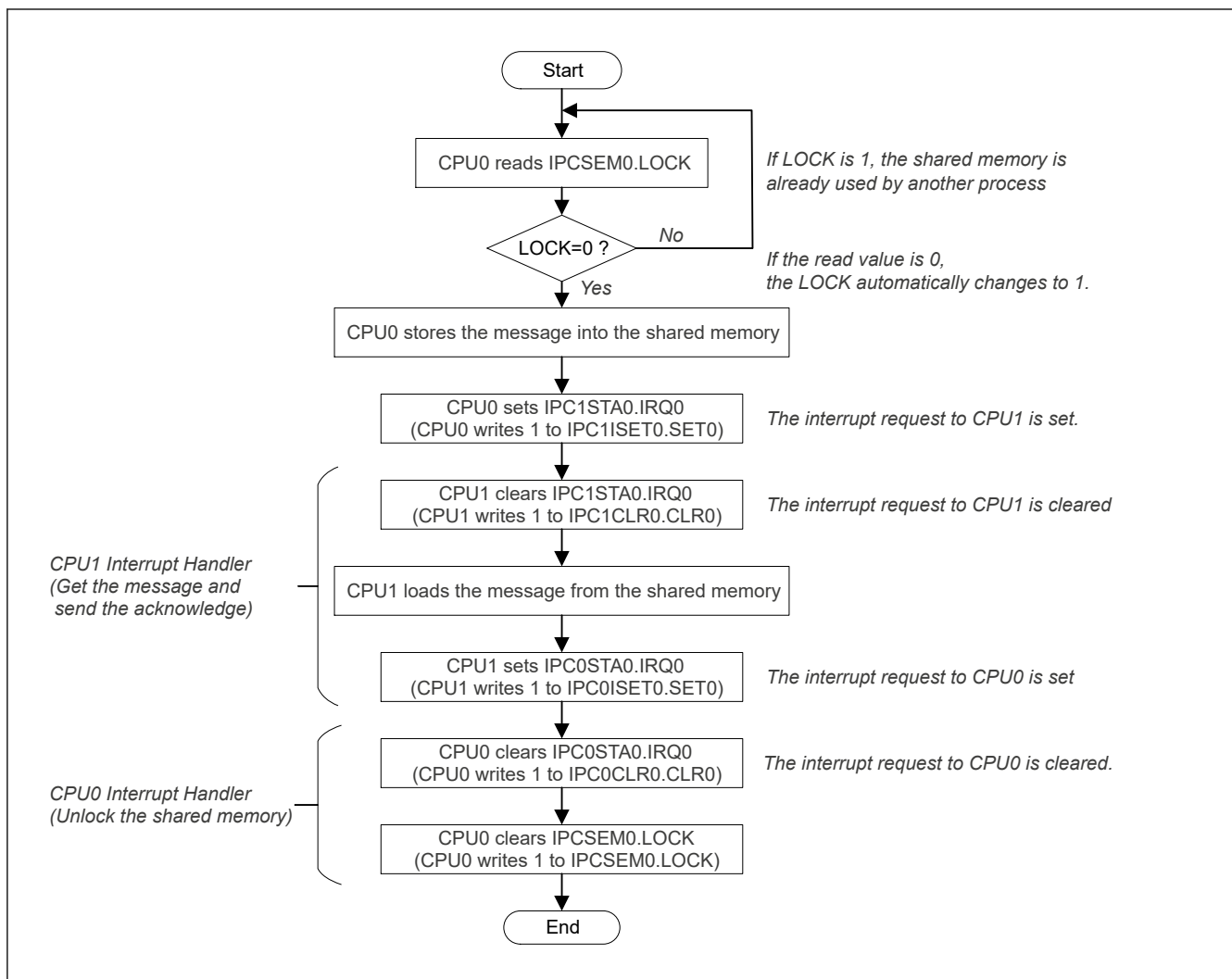


Figure 3.5 Example flow of CPU0 sending a message to CPU1

3.3.5 Message-FIFO

The IPC has four FIFOs. IPC00 and IPC01 are FIFOs that transmit data from CPU1 to CPU0, and IPC10 and IPC11 are FIFOs that transmit data from CPU0 to CPU1. Each FIFO issues the maskable interrupt $IPCiIRQj$ ($i = 0, 1$), ($j = 0, 1$) as shown in the following figure. $IPCiIRQj$ is issued when there is a request in the Message-FIFO ($RDY = 1$), when the Message-FIFO is full and you write to it ($FERR = 1$), or when the Message-FIFO is empty and you read from it ($RERR = 1$) in $IRQij$ ($i = 0, 1$; $j = 0, 1$). The security attributes of this register are controlled by IPCSAR. The privileged attribute of this register is controlled by IPCPAR. Security attributes and privileged attributes can be set for each FIFO. [Figure 3.6](#) shows an image diagram of the message-FIFOs.

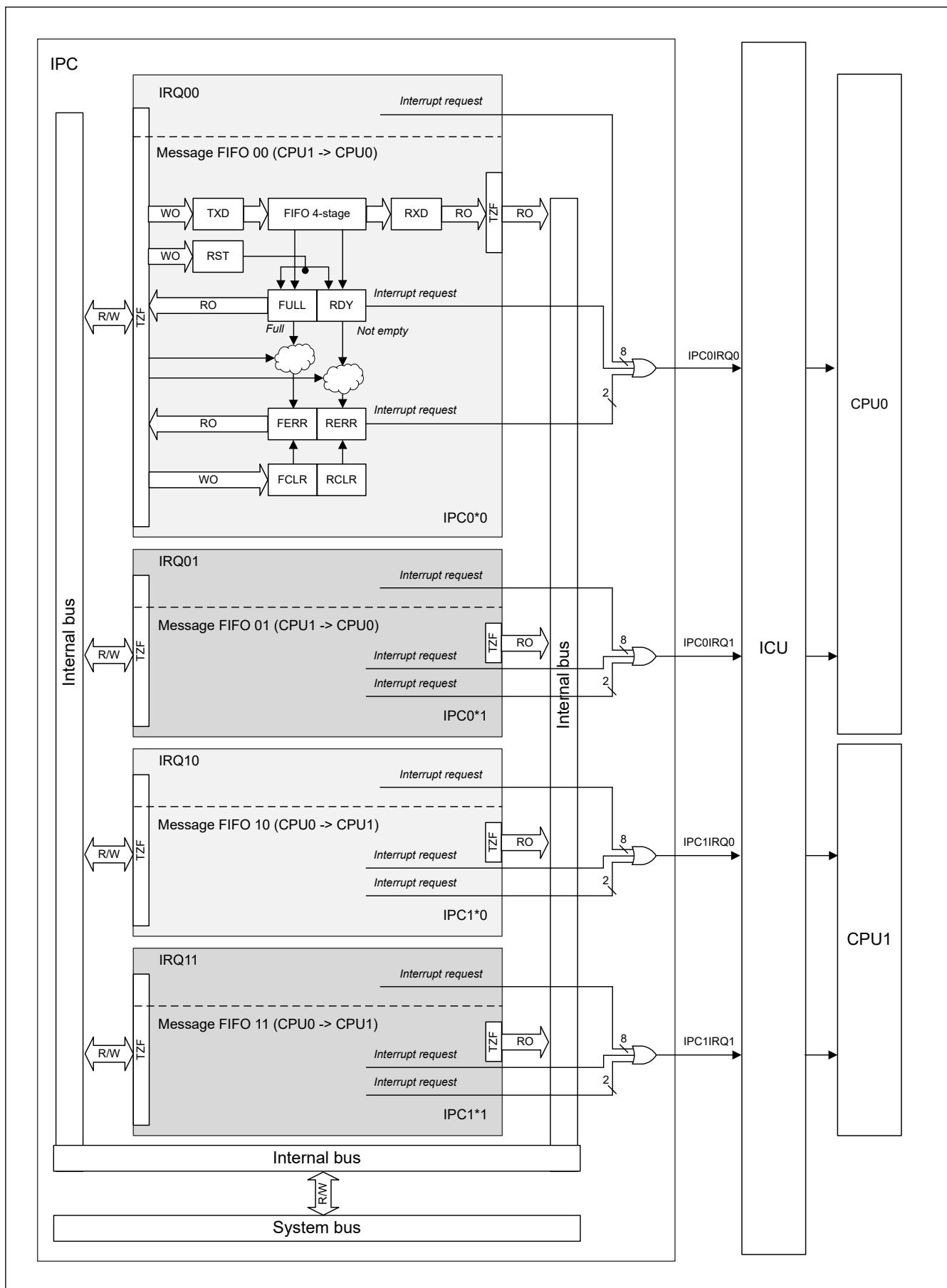


Figure 3.6 Inter-processor message FIFO

See Figure 3.4 for the security and privileged attributions.

Figure 3.7 shows an example of sending data from CPU1 to CPU0 using message FIFO 00.

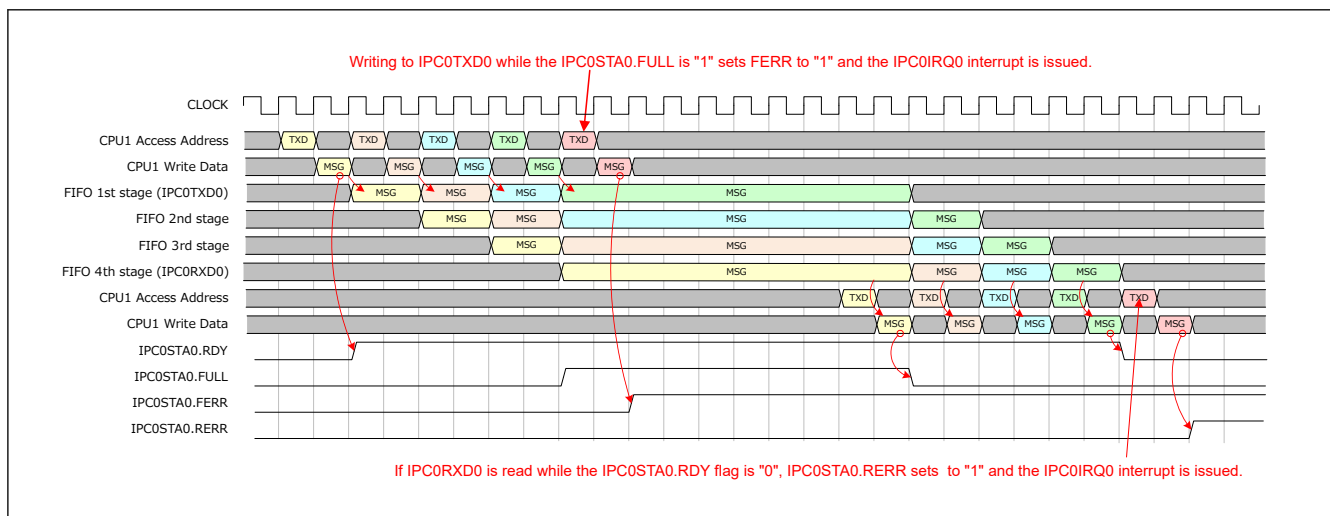


Figure 3.7 Inter-processor message FIFO timing

3.3.6 IPC Interrupt

CPU Sleep mode and CPU Deep Sleep mode can be canceled by the interrupts issued by IPC functions.